Practical 4: Time Series Structure and Correlation Analysis

Question: Using any dataset from the Time Series Data Library (TSDL), plot the time series and identify the presence of trend, seasonality, and irregular components. Also draw the ACF and PACF plots and interpret the nature of the series.

4.1 Objective

- To visualize a time series dataset and visually identify its structural components (Trend, Seasonality, and Noise).
- To decompose the series to separate these components effectively.
- To analyze the internal correlation structure of the series using Autocorrelation (ACF) and Partial Autocorrelation (PACF) plots.

4.2 Theory

- **COMPONENTS OF TIME SERIES:**
  - **Trend:** The long-term movement or direction of the data (e.g., increasing sales over years).
  - **Seasonality:** Repeating patterns or cycles that occur at fixed intervals (e.g., monthly or quarterly spikes).
  - **Irregular (Residual/Noise):** The random fluctuations that remain after trend and seasonality are removed.

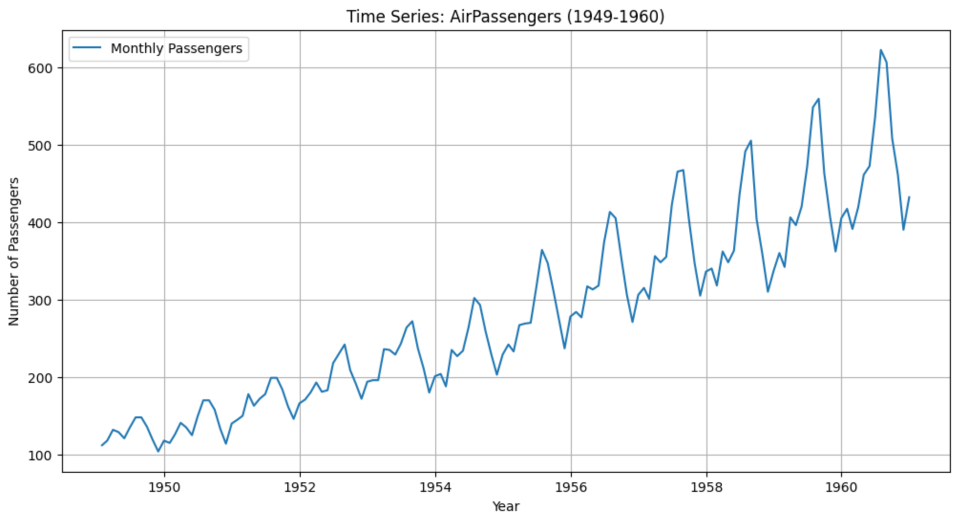
- **ACF (Autocorrelation Function):** Measures the correlation between the time series and a lagged version of itself. It helps identify Moving Average (MA) characteristics.

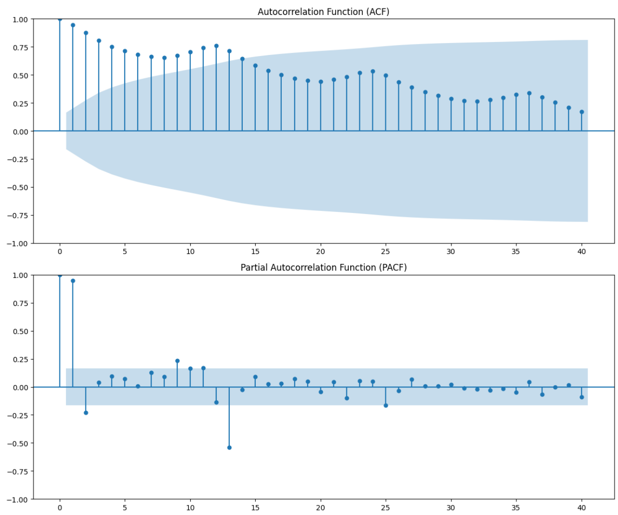
- **PACF (Partial Autocorrelation Function):** Measures the direct correlation between a specific lag and the current value, removing the effect of intermediate lags. It is useful for identifying AutoRegressive (AR) characteristics.

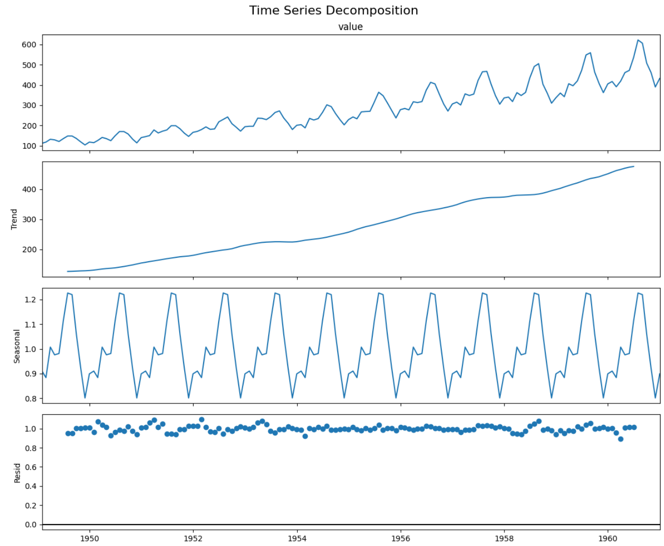
4.3 Code Implementation

```python
# 1. Load and Prepare Data
dataset = sm.datasets.get_rdataset("AirPassengers")
df = dataset.data
df['time'] = pd.date_range(start='1949-01-01', periods=len(df), freq='MS')
df.set_index('time', inplace=True)
ts_data = df['value']

# 2. Decompose the Series
# We use 'multiplicative' because the seasonal amplitude increases with the trend
decomposition = seasonal_decompose(ts_data, model='multiplicative')

# 3. Generate Correlation Plots
# These functions directly generate the ACF and PACF analysis plots
plot_acf(ts_data, lags=40)
plot_pacf(ts_data, lags=40)
```

4.3 Code Output







4.4 Interpretation & Conclusion

1. **Trend:** The decomposition plot clearly shows a steady upward trend, indicating the number of passengers is increasing over the years.
2. **Seasonality:** There is a distinct, repeating seasonal pattern every year (peaks in summer months), which confirms the data is seasonal.
3. **Model Choice:** Since the height of the seasonal waves increases as the trend goes up (the "funnel" shape in the original plot), a **Multiplicative Decomposition** is more suitable than an Additive one.
4. ACF/PACF:
   - The **ACF** plot shows a gradual decay, which is typical for a series with a trend. The "scalloped" wave shape in the ACF confirms the seasonality.
   - The **PACF** has a sharp cut-off after the first few lags, suggesting the immediate past values have a strong direct influence on the current value.